- **Internship Name : Data Analysis**
- **Task Name: Task 1: Python Setup, Jupyter Notebook, Variables, Data Types, and Operators**
- **Your Name : Shubhangi Naresh Khonde**
- **Internship Organization :**
- **Date:**

1. Objective

The objective of this task was to set up a Python development environment and practice the basic concepts of Python programming: variables, data types, arithmetic operators, and comparison operators.

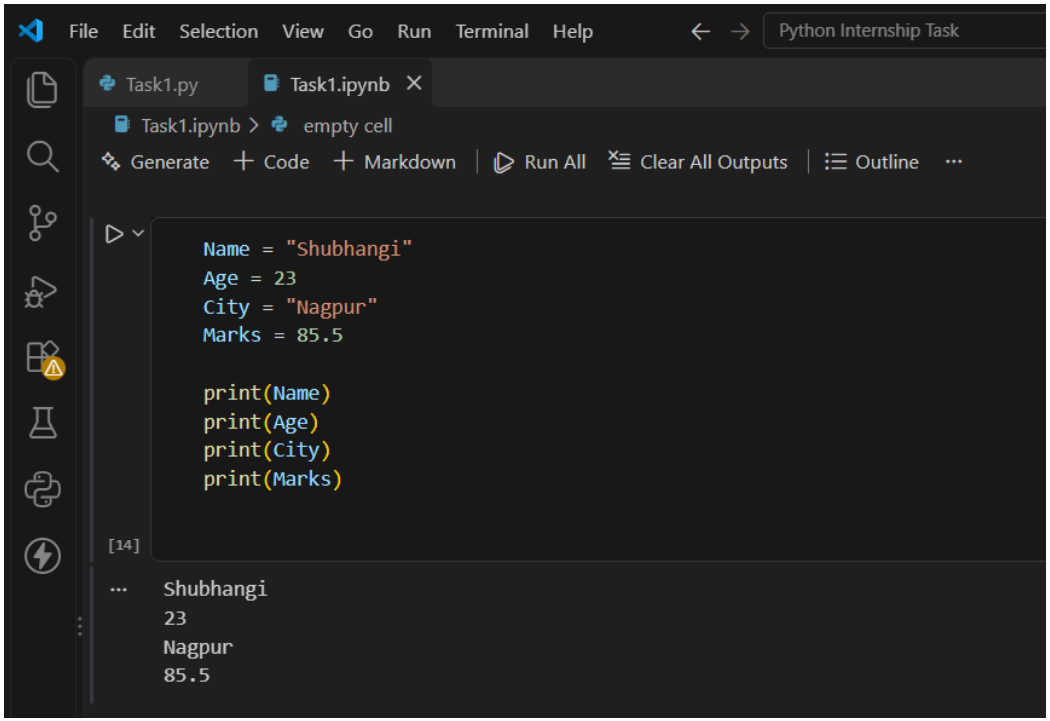
2. Tools Used

- **Python 3.13.2:** the programming language
- **Visual Studio Code (VS Code):** the code editor
- **Jupyter Notebook (.ipynb):** an interactive environment, run inside VS Code using the Python and Jupyter extensions

3. Environment Setup

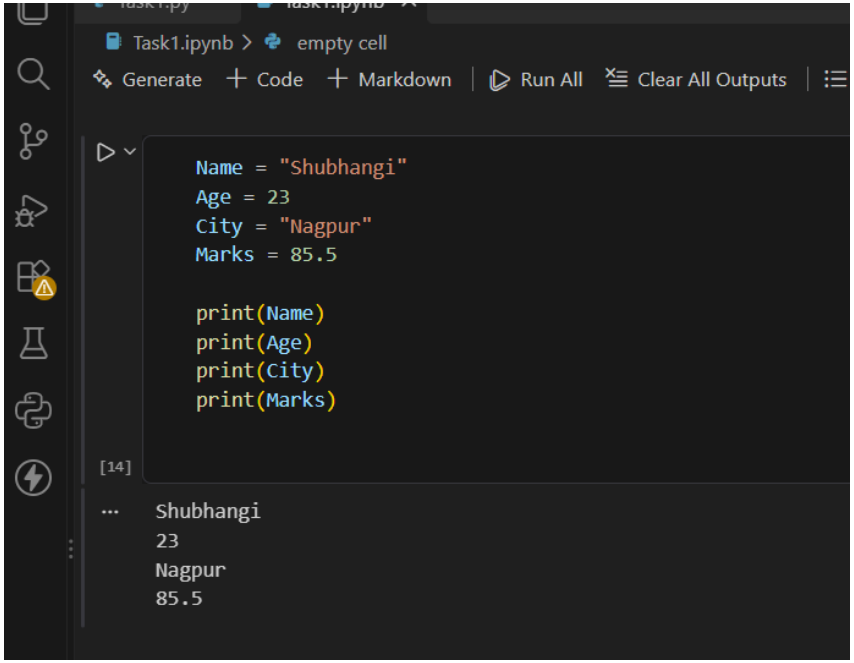
1. Installed Python on the system.
2. Installed Visual Studio Code.
3. Installed the **Python** and **Jupyter** extensions in VS Code.

4. Selected the Python 3.13.2 kernel and installed ipykernel.
5. Created two files, Task1.py and Task1.ipynb.



4. Variables

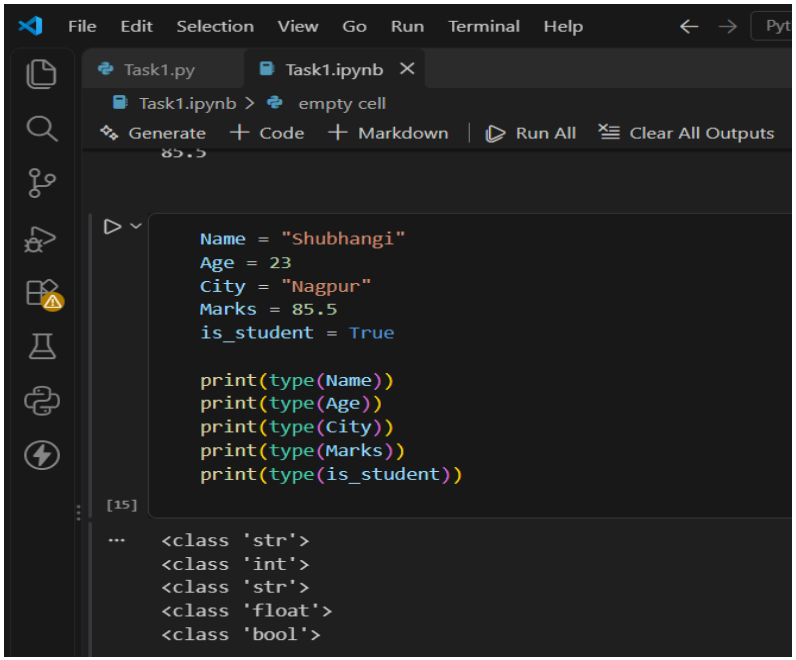
A variable is a named location used to store data in memory. Python does not need the data type to be declared, because it is detected automatically.



5. Data Types

Python has several built-in data types. The type() function is used to check the type of a value.

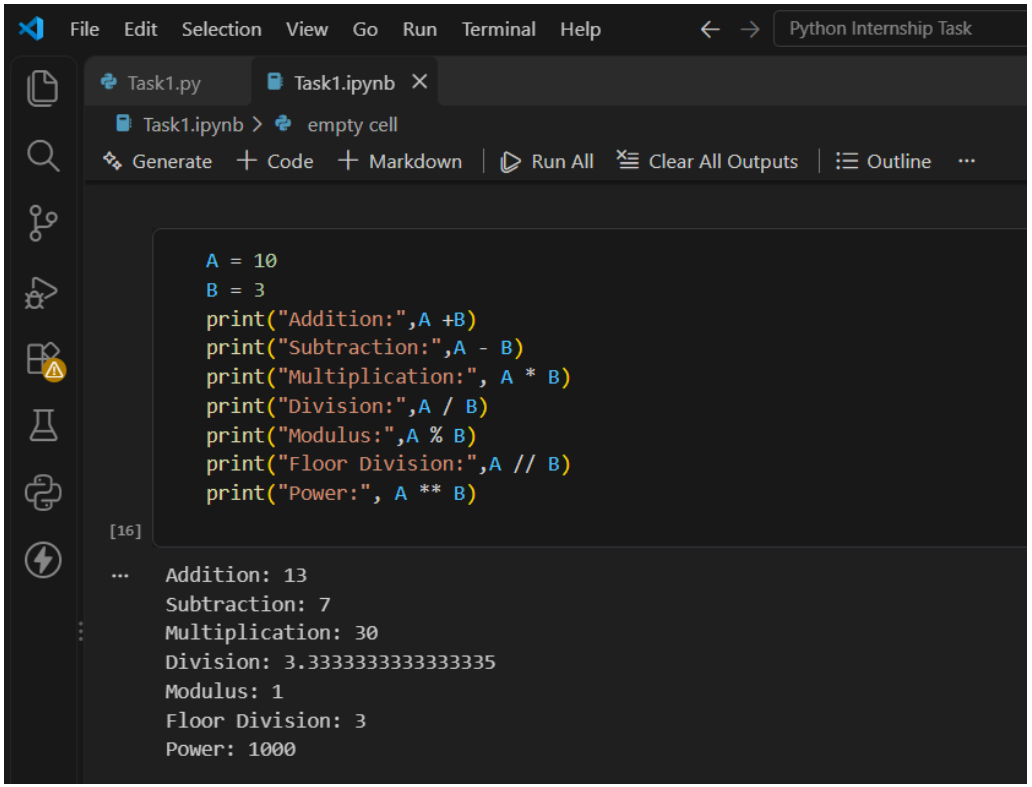
Data Type	Example	Description
str	"Shubhangi"	Text
int	23	Whole number
float	85.5	Decimal number
bool	True	True or False value



6. Arithmetic Operators

Arithmetic operators perform mathematical calculations. Here A = 10 and B = 3.

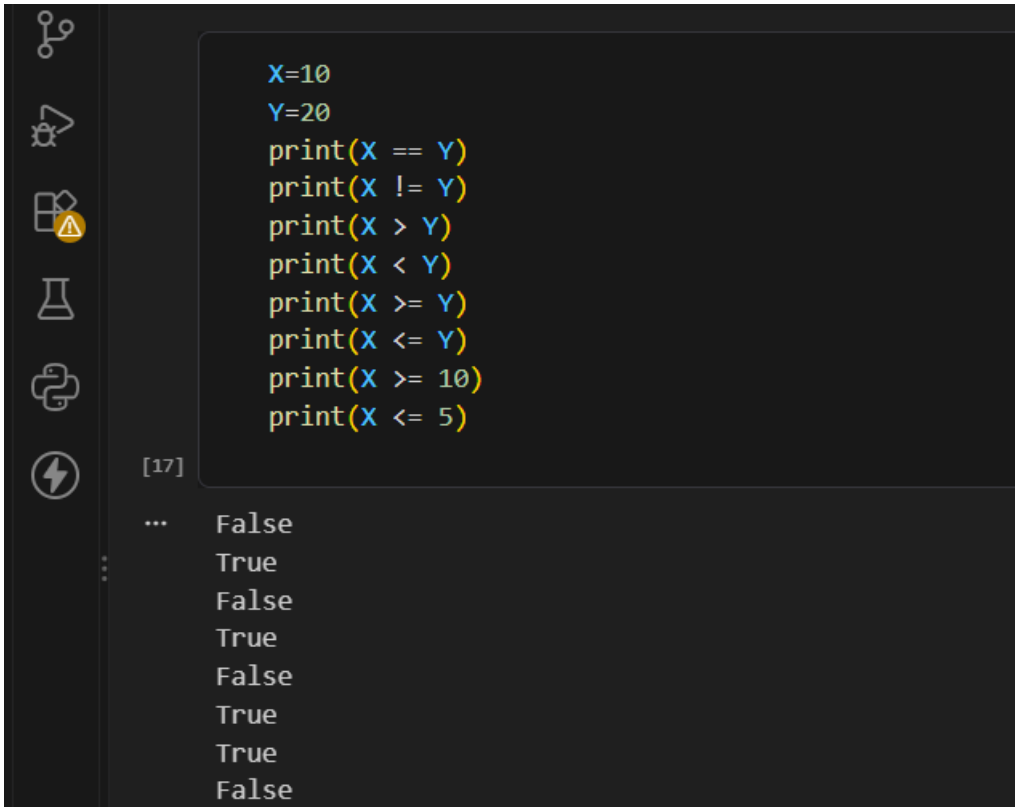
Operator	Operation	Result
+	Addition	13
-	Subtraction	7
*	Multiplication	30
/	Division	3.3333333333333335
%	Modulus (remainder)	1
//	Floor division	3
**	Power	1000


File Edit Selection View Go Run Terminal Help
Python Internship Task
Task1.py
Task1.ipynb
Task1.ipynb > empty cell
Generate
Code
Markdown
Run All
Clear All Outputs
Outline
A = 10
B = 3
print("Addition:",A +B)
print("Subtraction:",A - B)
print("Multiplication:", A * B)
print("Division:",A / B)
print("Modulus:",A % B)
print("Floor Division:",A // B)
print("Power:", A ** B)
[16]
Addition: 13
Subtraction: 7
Multiplication: 30
Division: 3.3333333333333335
Modulus: 1
Floor Division: 3
Power: 1000

7. Comparison Operators

Comparison operators compare two values and always return a Boolean result (True or False). Here X = 10 and Y = 20.

Expression	Meaning	Result
X == Y	Equal to	False
X != Y	Not equal to	True
X > Y	Greater than	False
X < Y	Less than	True
X >= Y	Greater than or equal to	False
X <= Y	Less than or equal to	True
X >= 10	Greater than or equal to 10	True
X <= 5	Less than or equal to 5	False



```
X=10
Y=20
print(X == Y)
print(X != Y)
print(X > Y)
print(X < Y)
print(X >= Y)
print(X <= Y)
print(X >= 10)
print(X <= 5)
```

[17]

```
False
True
False
True
False
True
True
False
```

8. Conclusion

In this task, I successfully set up Python and Jupyter Notebook in VS Code and practiced variables, data types, arithmetic operators, and comparison operators. I learned how to store data in variables, check data types using type(), perform calculations, and compare values using operators. This task gave me a strong foundation for learning more advanced Python topics such as conditions, loops, and functions.